13.3 The German Tank Problem

WORKSHEET

The question

A hostile naval power possesses a large, unknown number N of submarines. Interception of radio signals yields a small number n of their identification numbers X_i $(i = 1, 2, \ldots, n)$, which are taken to be independent and uniformly distributed over the continuous range from 0 to N. Show that Z_1 and Z_2, defined by

$$Z_1 = \frac{n+1}{n}\max\{X_1, X_2, \ldots, X_n\} \qquad \text{and} \qquad Z_2 = \frac{2}{n}\sum_{i=1}^{n} X_i,$$

both have means equal to N.

Calculate the variance of Z_1 and of Z_2. Which estimator do you prefer, and why?

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If M is the largest of n independent uniform values on $[0, N]$, what is $\mathrm{P}(M \le m)$?

STAGE 2 Their variances

Ask yourself Find $\mathrm{E}(M^2)$ and hence $\mathrm{Var}(M)$. For Z_2, what is the variance of a single uniform variable on $[0, N]$?

STAGE 3 Which is better?

Ask yourself Compare $\frac{1}{n(n+2)}$ with $\frac{1}{3n}$. How do they behave as n grows?

Try this next

1. **Mean squared error.** The biased estimator M has $\mathrm{E}(M-N)^2 = \mathrm{Var}(M) + \big(\mathrm{E}(M) - N\big)^2$. Compute it and compare with $\mathrm{Var}(Z_1)$. Find the constant c that minimises $\mathrm{E}(cM - N)^2$.

2. **The minimum too.** Show that $(n+1)\min\{X_i\}$ is also an unbiased estimator of N, and find its variance. Is it any good?

3. **The discrete formula.** For serial numbers $1, \dots, N$, sample k without replacement, with maximum m. Show that $\mathrm{P}(\max \le m) = \binom{m}{k} / \binom{N}{k}$, and verify that $m + \frac{m}{k} - 1$ is unbiased when $k = 1$.

4. **Tanks.** If 5 captured tanks have serial numbers 19, 40, 42, 60, 98, estimate the number of tanks made.

Further reading

- **Richard Ruggles & Henry Brodie, "An empirical approach to economic intelligence in World War II"**, *Journal of the American Statistical Association* 42 (1947). The original account.
- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on bias, variance and sufficiency.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican).

The full guided solution, and what the question leads to at university, is in Section 13.3:
https://beyondstep.keithmcnulty.org/ch13/german-tanks.html